

Group Members

1. Lim Chiew Ting
2. Loo Jie Qi
3. Ng Shen Zhi
4. Tan Huei Qing
5. Choo Jing Chuen

1. Describe Static Testing.
Test the system or test objects without running/executing the codes.
2. List and describe the Review Processes.
 1. Planning
 - Decide which documents in the software development process shall be subject to which review technique
 2. Kick off
 - Serves to provide those involved in the review with all of the necessary information
 3. Individual Preparation
 - The reviewers intensively study the review object and check it against the documents given as a basis for it as well as against their own experience.
 4. Review Meeting
 - A review leader or moderator leads the review meeting
 5. Rework
 - The manager decides whether to follow the recommendation or do something else
 - Usually, the author will eliminate the defects on the basis of the review results and rework the document.
 6. Follow up
 - The proper correction of defects must be followed up, usually by the manager, moderator, or someone especially assigned this responsibility.
3. Describe three advantages of Static Testing over Dynamic Testing?

Criteria	Static Testing	Dynamic Testing
Early Detection of Defects	Bugs and inconsistencies can be identified before execution, reducing rework	Defects are identified only after executing the code, leading to later detection
Cost Effective	Finding defects earlier in the lifecycle costs compared to later fixes	Fixing defects at later stage incurs higher costs due to debugging and rework
Improve Code Quality	Code reviews and analysis ensure adherence to coding standards and best practices	Focuses on runtime errors and performance, but may not ensure coding and standard.

4. Who are the participants in a Review and what are their roles?

Participants in a Review	Roles and responsibility
Author	Creates the document or code being reviewed. Clarifies doubt and incorporates feedback.
Moderator	Leads the review process, schedules meetings, ensures the review stays focused and productive.
Manager	Oversees the review process, ensures resources are available, and monitors progress.
Reviewers	Examine the document or code for defects, inconsistencies, and adherence to standards.

5. Imagine that you are the chairperson of a formal review, if there is conflict happening among the reviewers, how do you solve the conflicts between the reviewers?
Another review shall be conducted, to check on the areas that were not agreed upon. The end goal would be to come to a new agreement that is satisfactory to all reviewers. The process is repeated if another disagreement occurs, until all the conflicts are overcome.
6. List and explain the types of Review.
1. Code Review
 - Evaluate the code structure together with the tester.
 2. Desk Checking
 - Desk checking is the primary testing done on the code
 3. Pair Programming
 - Two programmers will collaborate together to write and test code. This type originates from Extreme Programming
 4. Pair Programming
 - Two programmers will collaborate together to write and test code. This type originates from Extreme Programming
 4. Code Walkthrough
 - The developer will present the code for evaluation of the code including the anomalies in a less formal meeting/
7. Analyse the following pseudo codes/codes and perform desk checking on the changes of variable's value:

Test Data for number is 2, 3, 1, 8, 10

```
BEGIN
Count = 0
Total = 0
REPEAT
Count = count + 1
Get number
Total = Total + number
UNTIL count = 5
Display Total
END
```